LLM-Primary Gold Annotation for Cyber Threat Intelligence Named Entity Recognition: Honest Evaluation of Token-Classification Architectures

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Abstract

Cyber threat intelligence (CTI) reports contain entity types with no general-domain analogue — ATT&CK technique identifiers, file hashes, TLP markings, threat actor aliases — that off-the-shelf NER models handle poorly. Prior work in this space, including an earlier iteration of this project, evaluates against *silver* labels produced by the same class of dictionary/rule annotator being improved upon, which we show is circular: a checkpoint that scores 0.74 F1 against dictionary-derived silver gold collapses to 0.35 against independently-produced gold, because the silver labels share the model's blind spots rather than measuring against them. We instead build gold directly with a frontier LLM (Claude Sonnet), which reads full advisory text and proposes exhaustive entity annotations under per-label span validators, followed by human curation on the evaluation split. We fine-tune three token-classification encoders — RoBERTa-base, RoBERTa-large (Liu et al., 2019), and SecureBERT (Aghaei et al., 2022), a RoBERTa-architecture model domain-pretrained on security text — on this gold and evaluate all three with an identical document-level harness, alongside GLiNER (Zaratiana et al., 2024), a zero-shot span-matching architecture, fine-tuned on the identical gold. SecureBERT is the best performer (0.71 strict micro-F1), ahead of RoBERTa-base (0.66) and RoBERTa-large (0.65), showing domain pretraining matters more than parameter count on this task; GLiNER trails all three token classifiers once retrained on the same honest gold (0.56). This reverses this project's own working hypothesis: an earlier comparison that scored an *un*retrained GLiNER checkpoint against retrained token classifiers found GLiNER ahead by 9–15 points and attributed it to architecture, a comparison this paper shows was not controlled for training data and does not hold once it is. All three token classifiers are near-perfect on dictionary-shaped structured types (attack technique id, CVE identifier, TLP classification, email address) and substantially weaker on semantic types with fewer, more varied training examples (threat actor group, hacking tool).

1 Introduction

Cyber threat intelligence reports — government advisories, malware analyses, incident write-ups — are the primary vehicle by which defenders share knowledge, yet extracting structured information from them remains largely manual. Named entity recognition over CTI text must handle entity types with no general-domain analogue: ATT&CK technique identifiers, file hashes, TLP markings, YARA rule names, threat actor aliases. Human annotation at scale is expensive in this domain because it requires security expertise, which motivated an earlier iteration of this project to avoid it entirely: build a *silver*-labeled corpus from agreement between a dictionary/regex annotator (STIX indicators, MITRE ATT&CK lookups) and zero-shot GLiNER (Zaratiana et al., 2024) predictions, and evaluate against held-out silver labels produced by the same dictionary annotator.

That evaluation is circular, and the gap it hides is large. A blind human audit of predicted spans found the dictionary annotator's own gold-recall was materially incomplete — it misses real entities the dictionary simply never encoded — so a model that learns to reproduce the dictionary's blind spots is rewarded, not penalized, by silver-label scoring. We confirm this quantitatively: rescoring an existing dictionary-silver-trained checkpoint against independently-produced gold drops its score from 0.74 to 0.35 F1 (§4). The model did not get worse; the measurement got honest.

This motivates the pipeline described here. Rather than avoiding human involvement entirely, we move it to where it is cheapest to apply correctly: a frontier LLM (Claude Sonnet) reads each advisory's full text and proposes exhaustive gold-standard entity annotations, subject to per-label span-format validators at generation time; a human

then curates the evaluation split (accepting, rejecting, or relabeling proposals), while the larger training split remains LLM-only. This is still not free of human effort, but it replaces exhaustive first-pass annotation with targeted review — and, critically, it produces evaluation gold that is *independent* of any dictionary or rule system a model might learn to imitate.

We ask four research questions:

RQ1 How large is the gap between circular (dictionary-silver) evaluation and honest (LLM-primary, human-curated) evaluation for a model trained on the old weak-supervision pipeline?

RQ2 Once both training and evaluation labels come from the honest LLM-gold pipeline, does model scale (RoBERTa-base vs. RoBERTa-large) or domain-adaptive pretraining (SecureBERT) matter more for token-classification CTI NER?

RQ3 Which entity types remain hard even after retraining on honest gold, and does the pattern track training-example frequency, structured-vs-semantic label type, or both?

RQ4 How does this fine-tuned token-classification fleet compare to a zero-shot span-matching architecture (GLiNER) retrained on the identical honest gold, and to a frontier LLM run zero-shot?

RQ1–RQ3 are answered here with data trained and scored on the current LLM-gold pipeline; so is RQ4's architecture half, with a GLiNER checkpoint retrained on the identical gold (§4.4). RQ4's frontier-LLM half remains **[PENDING:]**: the Gemini numbers from the prior weak-supervision iteration of this project are not comparable to the current gold and are not reported here.

2 Related Work

GLiNER (Zaratiana et al., 2024) treats NER as matching natural-language entity-type descriptions against learned span embeddings, enabling zero-shot extraction of arbitrary types; its schema-driven successor GLiNER2 (Zaratiana et al., 2025) unifies NER with classification and structured extraction in one model. Distant supervision for NER (Devlin et al., 2019) was formalized for open-domain text by BOND (Liang et al., 2020), which fine-tunes a token classifier on knowledge-base-derived labels then self-trains through a teacher-student loop to denoise them — the same failure modes we target with ensemble agreement (incomplete and noisy weak labels) motivate BOND's two-stage design.

Cybersecurity-specific NER corpora suffer acutely from schema fragmentation: Schmidt et al. (2025) show that naively unifying four CTI datasets under a coarse shared label set *hurts* performance relative to single-dataset training, while Ech-Chammakhy et al. (2025) demonstrate that principled, STIX-anchored harmonization recovers a 30% relative F1 gain over the same naive union — evidence that *how* labels are unified matters as much as how much data is unified; our LLM-primary annotation pipeline sidesteps unification entirely by defining the label schema once and generating gold directly for it, rather than merging pre-existing dictionary schemas. Deka et al. (2024) extend the entity-type inventory to attribution-specific concepts (impact, course-of-action) with deliberately long, sentence-level spans, and find instruction-fine-tuned LLMs outperform encoder taggers on their benchmark, motivating our use of an LLM as the *annotator* rather than the deployed extractor. Alam et al. (2024) benchmark LLMs directly on CTI tasks including entity extraction, providing the frontier-model baseline methodology RQ4's future-work extension will adapt.

LLM-as-annotator is itself an active line of work distinct from LLM-as-extractor: rather than running a frontier model zero-shot at inference time, we use it once, offline, to produce training and evaluation labels for smaller, cheaper, deployable encoders — a distillation-flavored setup, though unlike classic distillation the teacher's output here is explicitly checked (span-format validators, human curation) rather than trusted directly, because the failure mode we are guarding against is exactly the one BOND and the dictionary-silver approach share: an annotator's systematic blind spots being learned uncritically by the student.

3 Data and Method

3.1 Pipeline

We scrape approximately 5,000 CISA cybersecurity advisories; each is converted from PDF to markdown via docling, then stripped of layout artifacts and normalized (page-furniture dedup, pagination removal, split-hash rejoining) to a clean, model-facing text version. For 82 of these advi-

sories we generate gold entity annotations across 16 CTI-specific labels[1] directly with a frontier LLM (Claude Sonnet): the model reads each advisory's full clean text and proposes exhaustive entity spans, verbatim-matched back to the source text and rejected if they do not match exactly, and further gated by per-label span-format validators (hash length and charset, CVE/CWE/ATT&CK technique-ID grammar, IP/email format including defanged forms, TLP enumeration) so structurally malformed proposals never enter the gold set. This is an annotation-time generation gate, not a post-hoc filter.

The 66 training-split advisories receive LLM annotation only. The 16 validation/test advisories additionally go through human curation via a purpose-built review tool: a human accepts, rejects, or relabels each LLM proposal and may add entities the LLM missed, including an `UNK` category (not used for training or evaluation directly) for real entities that fit no current label, which we use to look for missing label types. Evaluation gold is therefore human-verified; training gold is not, which we report as a limitation (§5) but consider acceptable because the LLM-primary approach's failure mode — proposing an entity a human would reject — is structurally different from the dictionary approach's failure mode of *never proposing* entire classes of entity it has no rule for, which is what a human-verified evaluation set is positioned to catch either way.

The training split contains 6,701 gold entities across 66 advisories; the evaluation split (8 validation, 8 test advisories) contains 1,995 gold spans in the test set alone once chunked for scoring (§3.4). This is roughly 3.5× the raw entity volume of the prior dictionary-silver pipeline's test set on the same page count, because dictionary/regex annotation structurally omits several semantic label types (e.g. *affected software product*, *hacking tool*) that an LLM reader identifies from context.

3.2 Split

We split by advisory *document*, not by chunk, before chunking for training — 66 advisories train / 8 validation / 8 test — so that overlapping text windows from the same document cannot appear on both sides of the split. This was a deliberate fix carried over from the prior pipeline iteration: chunk-level splitting leaks document context (shared boilerplate, repeated entity mentions) across the train/test boundary and inflates apparent performance.

3.3 Models

RoBERTa-base and RoBERTa-large (Liu et al., 2019). Standard BIO token classifiers fine-tuned on the LLM-gold training split. Because gold spans are word-level and RoBERTa's tokenizer is subword-based, we tag the first wordpiece of each word and mask continuation wordpieces from the loss. Corpus chunks vary widely in length, and long or hash-heavy chunks fragment into more wordpieces than a single window admits under naive truncation; we instead tokenize with overlapping sliding windows (a 510-wordpiece budget per window, 50-word overlap between windows) so every word is covered by at least one window, and take each word's predicted tag from the window in which it sits most centrally at inference time. RoBERTa-base trains for 8 epochs at LR 2e-5, effective batch size 16; RoBERTa-large under the identical recipe with effective batch size 32 (gradient accumulation, to fit the larger model). Length-grouped batching reduces padding waste across the wide chunk-length range.

SecureBERT (Aghaei et al., 2022). The same RoBERTa architecture and fine-tuning recipe as RoBERTa-base (8 epochs, LR 2e-5, effective batch 16), but initialized from a checkpoint further pretrained on a security-text corpus rather than general-domain text. Since architecture and fine-tuning recipe are held identical to RoBERTa-base, any difference between the two isolates the effect of domain-adaptive pretraining from the effect of the fine-tuning data or procedure.

GLiNER (Zaratiana et al., 2024). `urchade/gliner_medium-v2.1`, zero-shot span-matching, fine-tuned on the identical LLM-gold training split: 5 epochs, LR 5e-5, effective batch 32 (batch 8, gradient accumulation 4), best checkpoint selected by validation loss (GLiNER's trainer reports no span-level F1 during training; that is computed separately with the same harness used for the token classifiers). Unlike the token classifiers, GLiNER handles its own tokenization and length limits internally rather than using the sliding-window scheme in §3.4; several of

[1] affected software product, attack technique id, attack technique name, cve identifier, cwe identifier, email address, file hash, file name, file path, hacking tool, ip address, malware family, tactic, threat actor group, tlp classification, yara rule name.

the longest test-set chunks were truncated at its 384-token limit, which we note as a limitation (§5) rather than paper over, since it plausibly disadvantages GLiNER specifically on labels that skew toward the end of long chunks.

Frontier LLM (future work). A zero-shot frontier-LLM baseline on the current gold, in the style of Alam et al. (2024), is deferred to future work; the Gemini numbers from the prior pipeline iteration are scored against dictionary-silver gold and are not comparable here.

3.4 Evaluation

All models are scored with the same document-level harness (`ner.evaluate`): predictions from overlapping chunks are merged back to document-global token indices before matching, so entities in chunk-overlap regions are not double-counted. We report strict (exact span-and-label match) and relaxed (label match plus span overlap) micro/macro precision, recall, and F1, on the 8-document test split, using the identical scoring implementation across models so the comparison in Section 4 is not an artifact of different matching rules.

4 Results

4.1 RQ1: Circular vs. Honest Evaluation

Table 1 rescores the prior pipeline's checkpoints — trained on dictionary-silver labels, previously reported at 0.74 and 0.76 F1 — against the current, independently-produced LLM+human gold, on the same 8 test advisories (1,995 gold spans) used throughout this paper. Both token-classification checkpoints collapse: RoBERTa-base from 0.740 to 0.351, SecureBERT from 0.755 to 0.393, roughly a 47–48% relative drop. The model did not get worse; the measurement got honest — these checkpoints were never evaluated against labels their own dictionary annotator didn't also produce, so silver scoring rewarded them for learning the dictionary's blind spots along with its correct patterns. For reference, the old GLiNER checkpoint (fine-tuned on the same dictionary silver labels, not yet retrained on the current gold) drops by a smaller margin, 0.666 to 0.540 (19% relative) — the asymmetry that motivated an earlier, informal version of this project's RQ4 hypothesis: that span-matching architectures generalize better than token classifiers once evaluation is honest. §4.4 retrains GLiNER on the identical gold and shows this smaller drop

Model	Circular F1	Honest P	Honest R	Honest F1
RoBERTa-base	0.740	0.86	0.22	0.351
SecureBERT	0.755	0.87	0.25	0.393
GLiNER (v1, ref.)	0.666	0.65	0.46	0.540

Table 1: Checkpoints trained on the prior dictionary-silver pipeline, scored against the silver labels they were trained toward ("Circular") vs. against independent LLM+human gold ("Honest"), strict span+label match, 8-document test split.

was not architectural after all; it reflects GLiNER's old training data happening to be less circular than the token classifiers' relative to *this specific honest eval set*, not an architecture effect that survives a controlled retrain.

4.2 RQ2: Scale vs. Domain Pretraining Under Honest Gold

Table 2 reports the three encoders after retraining on the LLM-gold training split and scoring against the same honest test gold. SecureBERT is the best performer (0.712 strict / 0.758 relaxed micro-F1), ahead of RoBERTa-base (0.659 / 0.683) and, notably, ahead of RoBERTa-large (0.648 / 0.697) despite RoBERTa-large having roughly 3× the parameters. Domain-adaptive pretraining recovers more honest-gold quality here than raw scale does. RoBERTa-large's higher relaxed-F1 and macro-F1 alongside a *lower* strict micro-F1 than RoBERTa-base is explained by the per-label pattern in §4.3: RoBERTa-large predicts the right general region and type more consistently across labels (helping macro and relaxed matching) but regresses sharply on the single highest-volume label, *malware family*, which drags support-weighted strict micro-F1 down.

Comparing back to Table 1: retraining on honest gold closes most of the gap the circular evaluation had hidden. RoBERTa-base's honest score rises from 0.351 (old checkpoint) to 0.659 (retrained) — most of the way back to its old, circular 0.740 — and SecureBERT's honest score (0.712) very nearly matches its old circular number (0.755). Fixing *both* the training labels and the evaluation labels recovers almost all of the quality the circular number implied was already there; fixing only the evaluation labels (Table 1) is what made the models look like they had regressed.

Once retrained on this same gold, GLiNER (0.56 strict / 0.58 relaxed) trails every token classifier, including RoBERTa-large. Its macro-F1 (0.62 strict)

Model	P	R	F1	F1 (relaxed)
RoBERTa-base	0.71	0.61	0.659	0.683
RoBERTa-large	0.71	0.60	0.648	0.697
SecureBERT	**0.73**	**0.70**	**0.712**	**0.758**
GLiNER	0.51	0.61	0.56	0.58

Table 2: Architecture comparison, identical LLM-gold training and evaluation labels, identical evaluation harness. Micro P/R/F1, 8-document test split, 1,995 gold spans.

sits well above its own micro-F1, the same pattern RoBERTa-large showed above — reasonably consistent type coverage dragged down by weak performance on a few high-volume labels, most strikingly *ip address* (F1 0.03, recall 0.02; §4.4 discusses why). This directly answers the question raised in §4's RQ1 discussion: the old GLiNER checkpoint's smaller circular-to-honest drop (Table 1) was not evidence of an architectural generalization advantage. Retrained under the identical procedure applied to the token classifiers, GLiNER is the weakest of the four models on this task.

4.3 RQ3: Which Entity Types Remain Hard

Table 3 breaks the three retrained models down by label. Structured, pattern-shaped labels are near-solved regardless of frequency: *attack technique id*, *cve identifier*, *tlp classification*, and *email address* all score 0.92–1.00 F1 across all three models, and even the lowest-support label in the test set (*cwe identifier*, $n = 4$) hits 1.00 on every model — though at that support level a single miss would swing the number by 0.25, so this particular result should be read as indicative rather than as evidence CWE identification is solved.

The weakest labels are not simply the rarest ones. *malware family* has the largest test-set support of any label ($n = 438$) yet tops out at 0.65 F1 (SecureBERT); *threat actor group* ($n = 157$) and *hacking tool* ($n = 139$) similarly lag well behind the structured labels despite moderate-to-high support. This suggests frequency alone does not explain the gap: these are open-vocabulary, context-dependent labels (whether a string names a piece of malware, a named intrusion set, or a legitimate admin tool like PowerShell often depends on surrounding sentence context rather than surface form), and that semantic ambiguity appears to be a harder problem for a token classifier than sparse supervision is. RoBERTa-large's regression on *malware family* specifically (0.35 vs. 0.59 for RoBERTa-base and 0.65 for SecureBERT, on identical training data) is the single largest per-label swing in the table and the main driver of RoBERTa-large's lower strict micro-F1 in Table 2.

Label	n	Base	Large	SecBERT
malware family	438	0.59	0.35	0.65
attack technique id	297	0.86	0.91	**0.99**
affected software product	159	0.48	0.57	0.61
threat actor group	157	0.29	0.42	0.34
attack technique name	144	0.57	0.53	0.57
hacking tool	139	0.37	0.44	0.51
ip address	130	0.85	0.85	0.92
tactic	111	0.59	0.63	0.62
file path	108	0.84	0.83	0.84
file hash	93	0.70	0.65	0.64
file name	89	0.76	0.81	0.74
cve identifier	55	0.92	0.96	0.94
tlp classification	36	0.97	0.97	0.97
email address	26	0.98	0.98	0.98
yara rule name	9	0.62	0.71	0.50
cwe identifier	4	1.00	1.00	1.00
Micro	1995	0.659	0.648	**0.712**

Table 3: Per-label strict F1, sorted by test-set support. Base = RoBERTa-base, Large = RoBERTa-large, SecBERT = SecureBERT.

4.4 RQ4: Architecture Comparison Resolved; Frontier Comparison Pending

GLiNER was fine-tuned on the identical LLM-gold training split used for the three token classifiers (5 epochs, LR 5e-5, effective batch 32; best checkpoint by validation loss) and scored with the identical harness on the identical 8-document test split. Table 2's result is unambiguous: GLiNER (0.56 strict F1) is the weakest of the four architectures tested, behind RoBERTa-large (0.65), RoBERTa-base (0.66), and SecureBERT (0.71). This resolves the question RQ1 raised and this project's own prior working hypothesis did not survive controlled retraining — §4 already sketches why: the old GLiNER checkpoint's smaller circular-to-honest drop reflected its old training data's relationship to *this* eval set, not a property of span-matching architecture that a fair retrain preserves.

Table 4 shows GLiNER is not uniformly weak — it matches or nearly matches the token classifiers on several structured labels (attack technique id 0.99, cwe identifier 1.00, cve identifier 0.96) — but is markedly worse on *file path* (0.42 vs. 0.83–0.84 for the token classifiers) and catastrophically worse on *ip address* (0.03 vs. 0.85–0.92), a label none of the three token classifiers found difficult. We do not have a confident explanation for the

Label	GLiNER F1	Best token clf. F1
ip address	0.03	0.92 (SecBERT)
file path	0.42	0.84 (Base/SecBERT)
threat actor group	0.29	0.42 (Large)
hacking tool	0.38	0.51 (SecBERT)
malware family	0.47	0.65 (SecBERT)
file name	0.46	0.81 (Large)
tactic	0.56	0.63 (Large)
attack technique name	0.50	0.57 (Base/SecBERT)
attack technique id	0.99	0.99 (SecBERT)
cve identifier	0.96	0.96 (Large)
cwe identifier	1.00	1.00 (all)

Table 4: GLiNER strict F1 vs. the best-performing token classifier per label (selected labels; full 16-label GLiNER results in `data/gold_v2/results/trained-gliner-v2.json`).

ip address collapse specifically; one contributing factor we can document rather than fully attribute is that GLiNER handles its own tokenization and truncates at a 384-token limit rather than using the token classifiers' sliding-window scheme (§3) — several of the longest test-set chunks were silently truncated during evaluation, which could plausibly cost recall on labels that cluster late in long chunks (IOC tables often appear near the end of CISA advisories), but we have not verified IP addresses specifically follow that pattern here, and flag this as a concrete follow-up rather than a settled explanation.

The frontier-LLM comparison remains pending: a zero-shot rerun on the current gold, in the style of Alam et al. (2024), has not been performed, and the prior Gemini numbers were scored against dictionary-silver gold and are not reported here to avoid repeating RQ1's circularity mistake.

5 Limitations

Training gold is LLM-only, not human-verified. Human curation is applied to the 16 validation/test advisories but not to the 66 training advisories (§3), for cost reasons. If Claude Sonnet's proposals have systematic blind spots of their own — the same structural risk that motivated moving off the dictionary annotator in the first place, just with a different annotator — training labels would inherit them even though the evaluation labels, being human-curated, would not. The results in §4 should be read as "honestly evaluated," not as "free of all annotator bias"; only the measurement side of that distinction is currently guaranteed.

Blind gold audit not yet complete. A blind, model-identity-hidden judgment sheet of 300 predicted spans (`paper/capstone/gold_audit.md`) is prepared but not yet human-scored. Once complete, it will report per-model precision independent of both the dictionary-silver noise that motivated this pipeline change and any residual bias in the LLM-primary gold itself, since a human judges each span directly against the source text rather than against either annotator's labels.

Test set size. Eight held-out advisories (1,995 gold spans in the test set) is larger than the prior pipeline's test set (1,442 dictionary-silver spans) and covers label types dictionary annotation structurally could not, but the lowest-support labels (*cwe identifier*, $n = 4$; *yara rule name*, $n = 9$) still have too few examples for their per-label numbers to be read as precise.

Windowing artifacts. All three token classifiers resolve each word's tag from whichever sliding window contains it most centrally at inference time, which is a heuristic, not a principled ensembling method; a small fraction of predictions near window boundaries may be affected.

GLiNER truncation, not windowing. Unlike the token classifiers, GLiNER's own tokenizer truncates rather than sliding-windows long input, and several of the longest test-set chunks were truncated at its 384-token limit during evaluation. We do not currently know how much of GLiNER's per-label weakness (§4.4), particularly the sharp *ip address* drop, is attributable to this versus a genuine architecture/data effect; disentangling the two — e.g. by re-chunking GLiNER's input at a shorter budget matching its limit, mirroring the token classifiers' windowing — is left to future work.

Frontier-LLM comparison (RQ4) incomplete. A zero-shot rerun on the current gold has not been performed (§4.4); the prior Gemini numbers were scored against the old dictionary-silver gold and are not comparable.

6 Conclusion

Evaluating against labels produced by the same annotator being improved upon is circular, and the gap it hides is not small: a checkpoint that scored 0.74–0.76 F1 against dictionary-silver gold scores 0.35–0.39 against independently-produced gold (RQ1). Retraining on that same honest gold

recovers most of the apparent quality — SecureBERT reaches 0.71 F1, nearly matching its own old circular number — showing the earlier result was not fabricated so much as measured against the wrong yardstick (RQ2). Domain-adaptive pretraining (SecureBERT) recovers more honest-gold quality than raw parameter count (RoBERTa-large), which notably regresses on the single highest-volume label rather than improving on it. Structured, pattern-shaped entity types are essentially solved by all three architectures; open-vocabulary, context-dependent types — most visibly *malware family* and *threat actor group* — remain the hard core of this task regardless of training volume (RQ3).

The architecture question this project set out to answer — whether GLiNER's span-matching design generalizes better than a token classifier's softmax head once measured honestly — is answered, and not in the direction this project's own earlier, uncontrolled comparison suggested. Retrained on the identical honest gold and scored with the identical harness, GLiNER (0.56 F1) is the weakest of the four architectures tested, behind every token classifier including RoBERTa-large (RQ4). The apparent GLiNER advantage in Table 1 was an artifact of comparing a retrained token classifier against an *un*retrained span-matching model — a reminder that the same circularity discipline this paper applies to silver-vs-honest gold (RQ1) has to be applied just as strictly to any two checkpoints being compared, or a training-data confound can masquerade as an architecture finding. **[PENDING: The frontier-LLM comparison and the blind gold-audit results (§5) remain outstanding and will be reported once available.]**

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